MTH1001

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

May 2018

Mathematical Structures

Module Leader: Professor Nigel Byott

Duration: 2 HOURS.

The mark for this module is calculated from 70% of the percentage mark for this paper plus 30% of the percentage mark for the January 2018 paper.

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) What does it mean for a sequence (a_n) of real numbers to **diverge to ∞**? (2)

   In each of the following cases, give an example of sequences (a_n) and (b_n) which both diverge to ∞, and which have the property stated:

   (i) $a_n - b_n \to \infty$;

   (ii) $a_n - b_n \to 1$;

   (iii) $a_n - b_n$ is bounded but divergent;

   (iv) a_n/b_n is a null sequence. (11)

   (b) Find the radius of convergence of each of the following power series (where $z \in \mathbb{C}$). Justify your answers.

   (i) $\displaystyle\sum_{n=1}^{\infty} \frac{z^n}{n+3i}$;

   (ii) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n z^n}{n^n}$;

   (iii) $\displaystyle\sum_{n=1}^{\infty} \frac{n! z^n}{\sqrt{n}}$. (12)

   (c) Let V be a vector space over $\mathbb{R}$.

   (i) State the definition of a linearly independent set in V.

   (ii) State the definition of a basis in V.

   (iii) Let $n \geq 2$, and suppose that the set $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ of n distinct vectors is a basis for V. For each of the following sets A, B, C, determine whether the set is linearly independent, is a basis in V, or is neither of these:

$$A = \{5\mathbf{v}_1, 5\mathbf{v}_2, \ldots, 5\mathbf{v}_n\};$$

$$B = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{n-1}\};$$

$$C = \left\{ \mathbf{v}_n, \mathbf{v}_{n-1}, \ldots, \mathbf{v}_2, \mathbf{v}_1 - \sum_{j=2}^{n} \mathbf{v}_j \right\}.$$

(15)

(d) Let

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 4 \\ -2 \\ -1 \end{pmatrix}, \quad \mathbf{v}_4 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Find a maximal linearly independent set from the vectors $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$, $\mathbf{v}_4$.

Determine whether the vector

$$\mathbf{w} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$$

can be written as a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$, $\mathbf{v}_4$ and, if it can, write $\mathbf{w}$ in this way. (10)

[**50**]

SECTION B

2. (a) (i) Give the definition of the identity function, id_A, on A. (3)

(ii) What does it mean for $f : A \to B$ to be invertible? (3)

(iii) Prove that if $f : A \to B$ is invertible, then f is bijective. (6)

(b) Give the definitions for a sequence (a_n) of complex numbers to

(i) converge to a limit $a \in \mathbb{C}$;

(ii) be bounded. (3)

Prove that any convergent sequence is bounded. (5)

Prove also that if (a_n) is a bounded sequence and (b_n) is a convergent sequence then the sequence $(a_n + b_n)$ is bounded. (5)

[**25**]

3. (a) In this question we write groups with multiplicative notation.

Give the definition of a *homomorphism* of groups. (2)

Let $\phi : G_1 \to G_2$ be a group homomorphism. If H is a subgroup of G_1 prove that $\phi(H) = \{\phi(a) : a \in H\}$ is a subgroup of G_2. (You may assume that homomorphisms preserve identity elements and inverses). (4)

Define

$$K = \{a \in G_1 : \phi(a) = e_2\}$$

(where e_j denotes the identity of G_j for $j = 1, 2$). Prove that K is a subgroup of G_1. Moreover, prove that if $a \in K$ and $c \in G_1$ then $cac^{-1} \in K$. (6)

(b) Let $\mathbb{R}^\infty$ be the $\mathbb{R}$-vector space of sequences (a_n) with entries in $\mathbb{R}$. Which of the following sets of sequences are subspaces of $\mathbb{R}^\infty$? Justify your answers.

(i) $U_1 = \{(a_n) \in \mathbb{R}^\infty : a_n = 0 \text{ when } n \text{ is a multiple of } 5\}$;

(ii) $U_2 = \{(a_n) \in \mathbb{R}^\infty : \lim_{n\to\infty} a_n^2 = 0\}$;

(iii) $U_3 = \{(a_n) \in \mathbb{R}^\infty : a_n \in \mathbb{Q} \text{ if } n \text{ is odd}\}$. (13)

[**25**]

4. (a) Let (a_n) be a sequence of real numbers. Determine whether each of the following statements is always true, giving a proof (if it is) or a counterexample (if it is not). Any standard results used should be clearly stated, but need not be proved.

   (i) If the sequence (a_n) converges to 0 then the series $\sum_{n=1}^{\infty} a_n$ converges.

   (ii) If the series $\sum_{n=1}^{\infty} a_n$ converges then the sequence (a_n^2) converges.

   (iii) If $a_n > 0$ and $a_{n+1} < a_n$ for all n, then the series $\sum_{n=1}^{\infty}(-1)^n a_n$ converges. (12)

   (b) Let U, V, W be vector spaces over $\mathbb{R}$

   (i) Give the definition of a linear map $f : V \to W$. (2)

   (ii) Suppose that $f : V \to W$ and $g : U \to V$ are linear maps. Prove that $f \circ g : U \to W$ is a linear map. (5)

   (iii) Assume that f has an inverse map $f^{-1} : W \to V$. Show that f^{-1} is a linear map. (6)

   [**25**]